

Finite Automata in Haskell

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Abstract

Finite state automata are popular tools to analyse formal languages. This project aims to implement finite state automata, both deterministic and nondeterministic. We also implement regular expressions and generate words from their language. Finally, we implement translations between our objects and run several tests to verify our implementation.

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1 Introduction

Finite state automata are mathematical models of computation. They are mostly used to study regular languages, i.e. languages with a certain structure. This structure is also captured by regular expressions.

In this report we discuss our Haskell-implementation of finite state automata and regular expressions. In particular, we implemented deterministic finite state automata, non-deterministic finite state automata and regular expressions. The goal is to make the equivalence of finite state automata and regular languages explicit, by various translation functions. Regular expressions are suitable for generating words from a regular language and in turn finite automata are suitable for evaluating if a word is in the language. Our implementation enables us to combine these features. Given a regular expression, we can compute another one that generates the complement language using finite state automata and given a regular expression, we can generate words that it accepts using regular expressions. Furthermore, we provide a toolkit that determines the minimal equivalent DFA to a DFA and checks isomorphism of DFA-s.

Our report is structured as follows. In Section 2 we give a brief introduction to the theory of regular languages and point towards more detailed references. In Sections 3 - 6 we discuss our implementation of finite state automata and regular expressions. In Section 7 we turn to the implementation of the translation functions. We then provide extensive testing apparatus for our implementation in Section 8 and conclude in Section 9.

2 Background Theory

Automata are mathematical models of computation. They are used in several areas of computer science and linguistics. Finite state automata, one of the most basic models, are used in, among other things, text processing and compilers. Here we will give a brief overview of the theory, which should be enough to make one understand the report. For a more detailed exposition, see for example [Sip13]. Automata are mathematical models of computation. They are used in several areas of computer science and linguistics. Finite state automata, one of the most basic models, are used in, for example, text processing and compilers.

2.1 Formal Languages

Automata are concerned with *formal languages*. A language in our setting is nothing more than a set of strings, or *words*, constructed from a given (finite) *alphabet*. An alphabet is typically denoted by the Greek capital Σ and consists of *symbols* or characters. A string over an alphabet is nothing more than a finite sequence of symbols. If A is a set, then the set of all finite sequences over A is denoted A^* . Hence A^* is the full language over alphabet A .

One special symbol that is a member of A^* for any A is ϵ . This character denotes the *empty string* and is just the word of length 0.

An example of an alphabet is $\Sigma = \{0, 1\}$ and one of the many possible languages over Σ is $B = \{0w \mid w \in \Sigma^*\}$. This language contains all binary strings starting with 0 and nothing more.

2.2 Combining Languages and Regular Expressions

We define several operations over languages to construct new ones. These are called the *regular operations*.

Let A and B be formal languages over alphabet Σ . Then:

- Their union, denoted as $A \cup B$, is defined as $\{w \mid w \in A \cup B\}$.
- Their concatenation, denoted as $A \circ B$ is defined as $\{wv \mid w \in A, v \in B\}$.
- We have already seen the closure operator $*$, called the *Kleene star*, to get $A^* := \{w_1w_2 \dots w_n \mid \forall i : w_i \in A, n \in \mathbb{N}\}$.
- Finally, we also have $A^+ := A^* \setminus \{\epsilon\}$ to get all non-empty sequences made with words from A .

These operations naturally give rise to *regular expressions*. A regular expression r always corresponds to a language, $\mathcal{L}(r)$.

Definition 2.1 (Regular Expressions). Let Σ be an alphabet. The set $\mathcal{R}$ of regular expressions over Σ are recursively defined as follows:

- $\emptyset \in \mathcal{R}$, denoting the empty language: $\mathcal{L}(\emptyset) = \emptyset$.
- $\epsilon \in \mathcal{R}$, denoting $\mathcal{L}(\epsilon) = \{\epsilon\}$.
- $a \in \mathcal{R}$, with $\mathcal{L}(a) = \{a\}$ for all $a \in \Sigma$.

If $r, r' \in \mathcal{R}$, then:

- $r + r' \in \mathcal{R}$, also written as $r|r'$, corresponding to $\mathcal{L}(r) \cup \mathcal{L}(r')$.
- $rr' \in \mathcal{R}$ with $\mathcal{L}(rr') = \mathcal{L}(r) \circ \mathcal{L}(r')$.
- r^* corresponding to $\mathcal{L}(r)^*$.
- r^+ with $\mathcal{L}(r^+)^+ = \mathcal{L}(r)^+$.

A regular expression r is like a pattern; any word that matches the pattern is *generated by* r . Languages generated by regular expressions are called *regular languages*.

2.3 Finite State Automata

As stated earlier, a finite state automaton (FSA) is a model of computation. Specifically, an FSA operates on words. When inputting a word into an FSA, it will either accept or reject. The set of all words that are accepted by an FSA A is called the language (recognized) by A . It is denoted $\mathcal{L}(A)$. We distinguish deterministic finite state automata (DFAs) from non-deterministic finite state automata (ϵ -NFAs). The formal definition of a DFA is as follows:

Definition 2.2. A DFA D is a 5-tuple $(S, \Sigma, \delta, s_0, F)$, where:

- S is a non-empty set of *states*.
- Σ is the alphabet on which D operates.
- $\delta : S \times \Sigma \rightarrow S$ is the transition function¹
- $s_0 \in S$ is the starting state.
- $F \subseteq S$ is the set of accepting states.

The computation of a word w on DFA $D = (S, \Sigma, \delta, s_0, F)$ will go as follows:

1. Let the current state s be s_0 .
2.
 - If $w = \epsilon$, accept if $s \in F$ and reject otherwise.
 - If $w = aw'$ where $a \in \Sigma$, let s be the next state $\delta(s, a)$ and let $w = w'$. Repeat step 2.

We have three remarks about this procedure:

- It is up for debate if a DFA rejects or fails when it reads a symbol that is not in the alphabet. Our implementation will reject for a more user-friendly experience.
- An NFA could end up in multiple states when reading a symbol. The idea is that the machine will non-deterministically choose to which state to go. If one possible path of computation ends up in an accepting state, the machine will accept. Note that some paths may end due to the machine reading a symbol a in a state s such that there is no t with $(s, a, t) \in \Delta$. In that case this path will count as not ending up in an accepting state.
- An ϵ -NFA extends the notion of an NFA by allowing for transitioning states on the empty symbol, namely ϵ . Before reading a symbol, the currently visited states need to be updated by all possible ϵ -transitions. This is done by taking the reflexive and transitive closure of the ϵ -relation of states.

Two FSAs are called *equivalent* just in case they recognize the same language. It may not be surprising that for every DFA there is an equivalent NFA: every function is a relation after all. However it turns out that for every NFA there is also an equivalent DFA. More on this in section 7.1. Moreover, every language that is accepted by an FSA is also generated by some regular expression and vice versa. I.e. the languages recognized by FSAs are exactly the regular ones. In the next sections we will discuss how the Haskell program shows these facts.

¹When considering an NFA, the only difference is that this is a transition relation $\Delta \subseteq S \times \Sigma \times S$.

3 DFA Implementation

This section describes our implementation of DFAs. As in the definition, a DFA consists of a tuple $(Q, \Sigma, \delta, q_{start}, A)$ representing the set of states, the alphabet, the transition function, the start state and the accept states. Throughout, we choose to represent sets as lists. The parameter of the DFA constructor stands for what object the states are. By default, we will use DFA-s with integer states, but occasionally we will need ones that have lists of integers as their states.

```
{-# LANGUAGE FlexibleInstances #-}
module DFA where
import Data.List ( (\\), intersect, nub )
import Test.QuickCheck ( elements, sublistOf, Arbitrary(arbitrary), Gen )

type Symbol = Char
data DFA a = DFA { states :: [a]
                  , alphabet :: [Symbol]
                  , delta :: Symbol -> a -> a
                  , start :: a
                  , acceptstate :: [a] }
```

We implement an evaluation function that upon input from the string, evaluates if the string is in the language. For this, we use a helper function `stateArr` that specifies the location after reading a list of symbols.

```
evaluate :: (Eq a, Ord a) => DFA a -> String -> Bool
evaluate df st = all ('elem' alphabet df) st && -- captures that all element of string in
  alphabet
  stateArr (start df) st 'elem' acceptstate df where
    stateArr q [] = q
    stateArr q (x:xs) = stateArr (delta df x q) xs
```

We write an example DFA on the alphabet $\Sigma = \{0,1\}$ accepting the language of words starting with a 0.

```
zeroStart :: DFA Int
zeroStart = DFA [0,1,2] ['0', '1'] deltazero 0 [1] where
  deltazero :: Symbol -> Int -> Int
  deltazero char st
    | st == 0 && char == '0' = 1
    | st == 0 && char == '1' = 2
    | st == 1 = 1
    | st == 2 = 2
    | otherwise = -1
```

We add a basic show instance for DFA-s. The transition function δ is shown as a list of strings specifying its value on the state space and the alphabet.

```
instance Show a => Show (DFA a) where
  show (DFA sts alph del strt acc)
    = "DFA" ++ "(" ++ show sts ++ "," ++ show alph ++ "," ++ show1 del ++ "," ++ show strt
      ++ "," ++ show acc ++ ")" where
    show1 f = show ["d" ++ "(" ++ show sym ++ "," ++ show st ++ ")"] ++ " = " ++ show (f
      sym st) | sym <- alph, st <- sts ]
```

On the DFA `zeroStart` from above, this yields the following string representation.

```
ghci> zeroStart
DFA([0,1,2],["01"],["d('0',0) = 1","d('0',1) = 1","d('0',2) = 2","d('1',0) = 2","d('1',1) = 1",
"d('1',2) = 2"],0,[1])
```

We write some functionalities for DFA-s that will be useful later. The `flipDFA` function changes the set of accept states in a DFA to its complement. The `cutDFA` function creates a DFA whose state space consists of words that are reachable from the start space via δ -transitions. The `reachables` function determines those reachable states. Using the `reachables` function, we can write a boolean function that determines if any words are accepted by the DFA.

```
flipDFA :: (Eq a, Ord a) => DFA a -> DFA a
flipDFA (DFA sts alp del st acc) = DFA sts alp del st (sts \\ acc)

reachables :: Eq a => DFA a -> [a]
reachables dfa = reachableInSteps [start dfa] where
  reachableInSteps ts | all ('elem' ts) (concatMap allSuccessors ts) = ts -- if all
    successors are in the set then
  --no more reachables
    | otherwise = reachableInSteps (nub $ ts ++ concatMap allSuccessors ts)
    --else add successors
  allSuccessors st = nub (st : [delta dfa sym st | sym <- alphabet dfa])

cutDFA :: Eq a => DFA a -> DFA a
cutDFA d = DFA sts alp delst strt acc where
  sts = reachables d
  alp = alphabet d
  strt = start d
  acc = reachables d 'intersect' acceptstate d
  delst = delta d

languageIsEmpty :: Eq a => DFA a -> Bool
languageIsEmpty d = null $ reachables d 'intersect' acceptstate d
```

For testing purposes, we add the functionality to be able to arbitrarily generate DFAs. For this we make DFAs an instance of `Arbitrary`. To generate arbitrary delta functions, we use two helper functions. The first generates arbitrary functions on the state space, the second adds the `Symbol` argument to the function. In the construction, we make sure that the state space and set of accept states are non-empty, by adding 0 to them. For running time purposes, we only generate DFA-s that have at most 4 states, but that can be adjusted as described in the code.

```
-- inspired by arbitrary instance in HW2
randomFunFromTo :: (Eq a, Ord a, Arbitrary a) => [a] -> [a] -> Gen (a -> a)
randomFunFromTo [] _ = return (const undefined)
randomFunFromTo (w:ws) ps = do
  f <- randomFunFromTo ws ps
  wResult <- elements ps
  return $ \v -> if v == w then wResult else f v

randomDelta :: (Eq a, Ord a, Arbitrary a) => [Symbol] -> [a] -> [a] -> Gen (Symbol -> a -> a)
randomDelta [] _ _ = return (const undefined)
randomDelta (sym:syms) ds ps = do
  f <- randomDelta syms ds ps
  wResult <- randomFunFromTo ds ps
  return $ \sy -> if sy == sym then wResult else f sy

instance Arbitrary (DFA Int) where
  arbitrary = do
    sts <- (0 :) <$> sublistOf [1..3] --choose a set up to 4 states. to change this,
      change 3 to larger
    let sym = ['0', '1']
    delst <- randomDelta sym sts sts
    strt <- elements sts
    accraw <- sublistOf sts
    let acc = nub (0:accraw)
    return $ DFA sts sym delst strt acc
```

4 NFA Implementation

This section describes our implementation of NFAs. In the initial design, the only (natural) difference to DFA-s is that the transition function maps to a list. The show instance of NFA-s is analogous to that of DFA-s.

```
{-# LANGUAGE FlexibleInstances #-}
module NFA where
import Data.List ( nub, union )
import DFA ( Symbol )
import Test.QuickCheck
  ( elements, sublistOf, Arbitrary(arbitrary), Gen )

data NFA a = NFA { statesNF :: [a]
                  , alphabetNF :: [Symbol]
                  , deltaNF :: Symbol -> a -> [a]
                  , startNF :: a
                  , acceptstateNF :: [a] }

instance Show a => Show (NFA a) where
  show (NFA sts alph del strt acc)
    = "NFA" ++ "(" ++ show sts ++ "," ++ show alph ++ "," ++ show1 del ++ "," ++ show strt
      ++ "," ++ show acc ++ ")" where
    show1 f = show ["d" ++ "(" ++ show sym ++ "," ++ show st ++ ")"] ++ " = " ++ show (f
      sym st) | sym <- alph, st <- sts ]
```

We also implement a similar, but more complicated evaluation function suitable for NFA-s. We have to keep track of all possible path we can be in via a helper function `stateArrNF`. We define the `unionL` helper function that takes the set union of elements of a list of lists. We make use of this functions on several other occasion throughout the code.

```
unionL :: Eq a => [[a]] -> [a]
unionL = foldr union []

evaluateNF :: Eq a => NFA a -> String -> Bool
evaluateNF nf st = all ('elem' alphabetNF nf) st && -- captures elements of string in
  alphabet
  any ('elem' acceptstateNF nf) (stateArrNF' [startNF nf] st) where
    stateArrNF' qs [] = qs -- type signature: [a]->Symbol-> [a]
    stateArrNF' [] _ = []
    stateArrNF' qs (x:xs) = unionL [stateArrNF' (deltaNF nf x q) xs | q <- qs]
      --recursion on string
```

We make NFA -s instance of `Arbitrary` by slightly modifying the relevant code for DFA-s. The difference is that the arbitrary functions we generate take lists as values.

```
-- recursively make a valuation function for these worlds:
randomFunFromTolist :: (Eq a, Arbitrary a) => [a] -> [a] -> Gen (a -> [a])
randomFunFromTolist [] _ = return (const undefined)
randomFunFromTolist (w:ws) ps = do
  f <- randomFunFromTolist ws ps
  wResult <- sublistOf ps
  return $ \v -> if v == w then wResult else f v

randomDeltaNF :: (Eq a, Arbitrary a) => [Symbol] -> [a] -> [a] -> Gen (Symbol -> a -> [a])
randomDeltaNF [] _ _ = return (const undefined)
randomDeltaNF (sym:syms) ds ps = do
  f <- randomDeltaNF syms ds ps
  wResult <- randomFunFromTolist ds ps
  return $ \sy -> if sy == sym then wResult else f sy

instance Arbitrary (NFA Int) where
  arbitrary = do
    -- choose a set of up to 10 worlds:
    sts <- (0 :) <$> sublistOf [1..5]
    let sym = ['0', '1']
```

```

delt <- randomDeltaNF sym sts sts
strt <- elements sts
accraw <- sublistOf sts
let acc = nub (0:accraw)
return $ NFA sts sym delt strt acc

```

5 Implementation of ϵ -NFA-s

Here we modify our implementation of NFA-s to account for epsilon transitions. We chose to represent ϵ transitions via a list of tuples of ϵ -related states. This is for convenience under taking ϵ -closure and to highlight the different role ϵ transitions take from standard symbol translation. The show instance is similar to DFA-s, with the ϵ -relations also listed, and is omitted here.

```

{-# LANGUAGE FlexibleInstances #-}
module ENFA where
import Data.List ( nub, sort )
import DFA ( Symbol )
import NFA ( randomDeltaNF, unionL )
import Test.QuickCheck ( elements, sublistOf, Arbitrary(arbitrary) )
data ENFA a = ENFA { statesENF :: [a]
                    , alphabetENF :: [Symbol]
                    , deltaENF :: Symbol -> a -> [a]
                    , epTrans :: [(a, a)]
                    , startENF :: a
                    , acceptstateENF :: [a]}

instance Show a => Show (ENFA a) where
  show (ENFA sts alph del eps strt acc)
    = "ENFA" ++ "(" ++ show sts ++ "," ++ show alph ++ "," ++ show1 del ++ "," ++ show eps
      ++ "," ++ show strt ++ "," ++ show acc ++ ")" where
    show1 f = show ["d" ++ "(" ++ show sym ++ "," ++ show st ++ ")"] ++ " = " ++ show (f
      sym st) | sym <- alph, st <- sts ]

```

To define ϵ -closure, we need some machinery to calculate reflexive and transitive closures. This work is inspired by [Jel13].

```

trClose :: Eq a => [(a, a)] -> [(a, a)]
trClose closure
  | closure == closureUntilNow = closure
  | otherwise                  = trClose closureUntilNow
  where closureUntilNow =
    nub $ closure ++ [(a, c) | (a, b) <- closure, (b', c) <- closure, b == b']

refClose :: Eq a => [a] -> [(a,a)] -> [(a,a)]
refClose as ps = nub (ps ++ [(x,x) | x <- as])

rtClose :: (Eq a, Ord a) => ENFA a -> [a] -> [a]
rtClose nf qs = sort (unionL [[p | p <- statesENF nf, (q, p) 'elem' trClose rcl] | q <- qs
  ]) where
  rcl = refClose (statesENF nf) (epTrans nf)

```

We use this to modify the evaluation function to account for ϵ -transitions.

```

evaluateENF :: (Eq a, Ord a) => ENFA a -> String -> Bool
evaluateENF nf st = all ('elem' alphabetENF nf) st && -- captures elements of string in
  alphabet
  any ('elem' acceptstateENF nf) (stateArrENF' [startENF nf] st) where
    stateArrENF' qs [] = rtClose nf qs
    stateArrENF' [] _ = []
    stateArrENF' qs (x:xs) =
      unionL ( [rtClose nf (stateArrENF' (unionL (map (deltaENF nf x) (rtClose
        nf [q]))) xs) | q <- qs ] )

```

```
-- this first does transitive closure on the input list, then transitive closure on
   output list and takes union
```

We make ENFA -s instance of Arbitrary by slightly modifying the relevant code for NFA-s to the effect of adding an arbitrary ϵ -relation list.

```
instance Arbitrary (ENFA Int) where
  arbitrary = do
    -- choose a set of up to 10 worlds:
    sts <- (0 :) <$> sublistOf [1..5]
    let sym = ['0', '1']
    delts <- randomDeltaNF sym sts sts
    eps <- sublistOf [(x, y) | x <- sts, y <- sts]
    strt <- elements sts
    accraw <- sublistOf sts
    let acc = nub (0:accraw)
    return $ ENFA sts sym delts eps strt acc
```

6 Regular expressions

In this section, we discuss the implementation of regular expressions, as well as a small toolbox to manipulate them.

```
module RegExp where
import DFA
import NFA()
import ENFA()
import Test.QuickCheck
```

6.1 Definition

Recall that a regular expression is defined to be either empty, the emptystring, a single symbol, the union or concatenation of two expressions, or the closure of a language (with or without the emptystring). The datatype **RegExp** we used closely follows this definition, but also allows for some more flexibility.

```
data RegExp = Empty | Epsilon | R [Symbol] | Union RegExp RegExp | Con RegExp RegExp | Star
             RegExp | Plus RegExp
  deriving (Show, Eq)
```

The main difference between **RegExp** and regular expressions, is in the base case for single symbols in the original definition. We chose to allow a list of symbol to increase user-friendliness. This means that instead of **Con** (R "a") (R "b"), one can write **R** "ab" to denote the same expression. **R []** is also equivalent to **Epsilon**.

For convenience, we also added a function that takes a list of regular expressions and outputs the expression corresponding to the union of all its members.

```
regExpUnion :: [RegExp] -> RegExp
regExpUnion = simplify . foldr Union Empty
```

6.2 Printing

As larger expressions tend to become unreadable, we implemented a function for pretty showing and pretty printing regular expressions.

```
pRegExp :: RegExp -> String
pRegExp Empty = ""
pRegExp Epsilon = "R e"
pRegExp (R []) = "R e"
pRegExp (R xs) = show xs
pRegExp (Union r1 r2) = "(" ++ pRegExp r1 ++ "|" ++ pRegExp r2 ++ ")"
pRegExp (Star r) = "(" ++ pRegExp r ++ ")*"
pRegExp (Con r1 r2) = pRegExp r1 ++ pRegExp r2
pRegExp (Plus r) = "(" ++ pRegExp r ++ "+)"

ppRegExp :: RegExp -> IO ()
ppRegExp = print . pRegExp
```

Here is an example to illustrate the improvement on the readability.

`ppRegExp (Con (Union (Con (R "5") (R "3")) (R [])) (Star (R "19")))` yields
`("53"|R e)("19")*`.

6.3 Arbitrary Generation

We create an Arbitrary instance for `RegExp` in order to be able to generate regular expressions. The main purpose for this is testing of properties. Note that we exclude the `Empty` expression from the generation, as empty expressions cannot generate words. The parameter for `sized` decreases swiftly in order to keep the generated expressions of a reasonable and readable size. This is necessary for feasible runtimes in the translation process from regular expressions to DFAs later on.

```
instance Arbitrary RegExp where
  arbitrary = sized randomReg where
    randomReg :: Int -> Gen RegExp
    randomReg 0 = R <$> elements ["0", "1"]
    randomReg n = oneof [ R <$> elements ["0", "1"]
                        , Star <$> randomReg (n `div` 8)
                        , Plus <$> randomReg (n `div` 8)
                        , Con <$> randomReg (n `div` 8)
                        , <*> randomReg (n `div` 8)
                        , Union <$> randomReg (n `div` 8)
                        , <*> randomReg (n `div` 8)
                        , return Epsilon ]
```

```
generateString :: RegExp -> Gen String
generateString = ('generateString' 5) . simplify
  where generateString' :: RegExp -> Int -> Gen String
        generateString' Empty _ = error "cannot generate from empty language"
        generateString' Epsilon _ = return ""
        generateString' (R xs) _ = return xs
        generateString' (Union r1 r2) n = oneof [generateString' r1 n, generateString' r2 n]
        generateString' (Con r1 r2) n = do
          w <- generateString' r1 n
          v <- generateString' r2 n
          return (w ++ v)
        generateString' (Star _) 0 = return ""
        generateString' (Star r) n = oneof [generateString' Epsilon n, generateString' (Con r (Star r)) (n - 1)]
        generateString' (Plus r) n = generateString' (Con r (Star r)) n
```

```

simplify :: RegExp -> RegExp
simplify r | r == simplify' r = r
           | otherwise = simplify $ simplify' r

where
simplify' :: RegExp -> RegExp
simplify' Empty = Empty
simplify' Epsilon = Epsilon
simplify' (R xs) = R xs
simplify' (Con Empty Empty) = Empty
simplify' (Con Epsilon Epsilon) = Epsilon
simplify' (Con _ Empty) = Empty
simplify' (Con Empty _) = Empty
simplify' (Con r' Epsilon) = simplify' r'
simplify' (Con Epsilon r') = simplify' r'
simplify' (Con r1 r2) = Con (simplify' r1) (simplify' r2)
simplify' (Union Empty Empty) = Empty
simplify' (Union Empty r') = simplify' r'
simplify' (Union r' Empty) = simplify' r'
simplify' (Union Epsilon Epsilon) = Epsilon
simplify' (Union r1 r2) | r1 /= r2 = Union (simplify' r1) (simplify' r2)
                        | otherwise = simplify' r1
simplify' (Plus Epsilon) = Epsilon
simplify' (Star Epsilon) = Epsilon
simplify' (Plus Empty) = Empty
simplify' (Star Empty) = Epsilon
simplify' (Star r') = Star (simplify' r')
simplify' (Plus r') = Plus (simplify' r')

```

7 Translation functions

Here we implement all the algorithms that translate between FSAs and regular expressions. For each instance, we give a short theoretical background and then discuss our implementation.

```

module Translation where
import Data.List
import NFA
import DFA
import ENFA
import RegExp
import Data.Maybe (fromJust)

```

7.1 The Powerset construction

As stated earlier, (ϵ) -NFAs and DFA-s are equivalent in their computing power. This fact is shown by constructing an equivalent DFA from an (ϵ) - NFA. This procedure is called the powerset construction, which we detail here.

Definition 7.1. (Powerset construction - NFA-s) Given an NFA $N = (Q_n, \Sigma, \delta_n, q_0, F_n)$, its powerset construct is a DFA $D = (\mathcal{P}(Q_n), \Sigma, \delta, S, F_D)$, where

- $\mathcal{P}(Q_n)$ is the powerset of Q_n ,
- $S = \{q_0\}$
- $F_D = \{R \subset Q_N : S \cap F_n \neq \emptyset\}$

- $\delta_D(R, \sigma) = \cup_{s \in R} \delta_N(s, \sigma)$

For ϵ -NFA-s the powerset construct is defined similarly, with the distinction that $S = \text{ecl}(\{q_0\})$ and $\delta_D(R, \sigma) = \text{ecl}(\cup_{s \in R} \delta_N(s, \sigma))$. Here, $\text{ecl}(A)$ stands for the set of states reachable from A via ϵ -transitions.

Theorem 1. *An (ϵ)-NFA N and its powerset construct are equivalent.*

We implement the powerset construction for NFA-s. The states of the powerset construct are sublists of states of the NFA. Note that the standard powerset construction is greedy: it introduces exponentially many states. For running purposes, we cut the resulting DFA by our `cutDFA` function. In some examples, this could cut down a DFA on 2^{19} states to as few as 10.

```
nfaToDFA :: (Eq a, Ord a) => NFA a -> DFA [a]
nfaToDFA (NFA sts alph del strt ac) =
  cutDFA $ DFA sortedStates alph del' (sort [strt]) [st | st <- sortedStates, intersect st
    ac /= []] where
    del' sy ls = unionL [del sy l | l <- ls]
    sortedStates = map sort $ subsequences sts
```

For ϵ -NFA-s, we need to take care to take ϵ -closures at each relevant step.

```
enfaToDFA :: (Eq a, Ord a) => ENFA a -> DFA [a]
enfaToDFA (ENFA sts alph del eps strt ac) = let nf = ENFA sts alph del eps strt ac in
  cutDFA $ DFA sortedStates alph del' (rtClose nf (sort [strt])) [st | st <- sortedStates,
    intersect st ac /= []] where
    del' sy ls = let nf = ENFA sts alph del eps strt ac in rtClose nf ( unionL [del sy l |
      l <- ls] )
    sortedStates = map sort $ subsequences sts
```

7.2 From regular expressions to ϵ -NFAs

Now we implement transition from `RegExp` to `ENFA`.

```
-- RegExp to ENFA
regExpToENFA :: RegExp -> ENFA Int
regExpToENFA Empty = ENFA [1] [] (\_ _ -> []) [] 1 []
regExpToENFA Epsilon = ENFA [1] [] (\_ _ -> []) [] 1 [1]
regExpToENFA (R []) = ENFA [1] [] (\_ _ -> []) [] 1 [1]
regExpToENFA (R xs) = ENFA [0..length xs] (nub xs) delta2 [] 0 [length xs] where
  delta2 symbol state | state >= length xs || state < 0 = []
    | xs !! state == symbol = [state + 1]
    | otherwise = []
regExpToENFA (Union r1 r2) = regExpToENFA r1 'unionENFA' makeDisjoint (regExpToENFA r1) (
  regExpToENFA r2)
regExpToENFA (Star r) = starENFA (regExpToENFA r)
regExpToENFA (Con r1 r2) = regExpToENFA r1 'concatENFA' makeDisjoint (regExpToENFA r1) (
  regExpToENFA r2)
regExpToENFA (Plus r) = regExpToENFA r 'concatENFA' makeDisjoint (regExpToENFA r) (starENFA
  (regExpToENFA r))

-- function that takes two ENFAs and outputs a relabeling of the second ENFA such that the
  states of both become disjoint
makeDisjoint :: ENFA Int -> ENFA Int -> ENFA Int
makeDisjoint n1 n2 = ENFA states' alphabet' delta' epT start' accept where
  add | minimum (statesENF n2) <= 0 = negate (minimum (statesENF n2)) + (maximum (statesENF
    n1 ++ statesENF n2) + 1)
    | otherwise = maximum (statesENF n1 ++ statesENF n2) + 1
  states' = map (+ add) (statesENF n2)
  alphabet' = alphabetENF n2
```

```

delta' sym state = map (+ add) (deltaENF n2 sym (state - add))
epT = [(s + add, t + add) | (s,t) <- epTrans n2 ]
start' = startENF n2 + add
accept = map (+ add) (acceptstateENF n2)

unionENFA :: ENFA Int -> ENFA Int -> ENFA Int -- Use only if states are disjoint
unionENFA n1 n2 = ENFA states'' alphabet2 delta'' epT start'' accept where
  states'' = start'' : statesENF n1 ++ statesENF n2
  alphabet2 = alphabetENF n1 'union' alphabetENF n2
  delta'' sym st = deltaENF n1 sym st ++ deltaENF n2 sym st
  epT = epTrans n1 ++ epTrans n2 ++ [(start'', startENF n1), (start'', startENF n2)]
  start'' = maximum (statesENF n2 ++ statesENF n1) + 1
  accept = acceptstateENF n1 ++ acceptstateENF n2

starENFA :: ENFA Int -> ENFA Int
starENFA n = ENFA (statesENF n) (alphabetENF n) (deltaENF n) ep (startENF n) (
  acceptstateENF n) where
  ep = epTrans n ++ [(startENF n, s) | s <- acceptstateENF n] ++ [(s, startENF n) | s <-
    acceptstateENF n]

concatENFA :: ENFA Int -> ENFA Int -> ENFA Int -- Use only if states are disjoint again
concatENFA n1 n2 = ENFA states4 alphabet'' delta3 epT start3 accept where
  states4 = statesENF n1 ++ statesENF n2
  alphabet'' = alphabetENF n1 'union' alphabetENF n2
  delta3 sym st = deltaENF n1 sym st ++ deltaENF n2 sym st
  epT = epTrans n1 ++ epTrans n2 ++ [(s, startENF n2) | s <- acceptstateENF n1]
  start3 = startENF n1
  accept = acceptstateENF n2

```

Finally, we turn to implement DFA to RegExp. AFL-notes version:

First we rename the states so that they are now integers $1 \dots n$ where n is the amount of states. For that we use the following function:

```

makeIntDFA :: (Eq a, Ord a) => DFA a -> DFA Int
makeIntDFA dfa = DFA sts alph delt strt acceptst
  where sts = [1..length (states dfa)]
        alph = alphabet dfa
        delt sym i = indX $ delta dfa sym (states dfa !! (i-1))
        strt = indX (start dfa)
        acceptst = [indX s | s <- acceptstate dfa]
        indX s = fromJust (elemIndex s (states dfa)) + 1

-- simplify to make the result a bit more readable
dfaToRegExp :: Ord a => DFA a -> RegExp
dfaToRegExp dfa = simplify $ regExpUnion [rijk dfaInt (start dfaInt) f (length $ states
  dfaInt) | f <- acceptstate dfaInt ]
  where dfaInt = (makeIntDFA . minimizedDFA) dfa

-- Here is the magic from the notes:
rijk :: DFA Int -> Int -> Int -> Int -> RegExp
rijk dfa i j 0 | i == j = regExpUnion $ Epsilon : labels
  | null labels = Empty
  | length labels == 1 = head labels
  | otherwise = foldr Union (head labels) (tail labels)
  where labels = [R [x] | x <- alphabet dfa, delta dfa x i == j]
rijk dfa i j k = Union (rijk dfa i j (k-1)) (Con (rijk dfa i k (k-1)) (Con (Star $ rijk dfa
  k k (k-1)) (rijk dfa k j (k-1))))

```

Brzowski's algorithm for minimising DFA's [Brz62]

```

minimizedDFA :: Ord a => DFA a -> DFA [Int]
minimizedDFA df | all ('elem' acceptstate df) (reachables df) = DFA [[0]] (alphabet df) (
  const (const [0])) [0] [[0]]

```

```

        | otherwise = (reverseDFA . reverseDFA) df

reverseDFA :: Ord a => DFA a -> DFA [Int]
reverseDFA d = enfaToDFA' $ ENFA sts alph delt ep st acc where
    e = singleAccept $ cutDFA d
    sts = statesENF e
    alph = alphabetENF e
    delt sym s = [t | t <- statesENF e, s `elem` deltaENF e sym t]
    ep = [(s, t) | (t, s) <- epTrans e]
    st = head $ acceptstateENF e
    acc = [startENF e]
    enfaToDFA' nf@(ENFA sts' alph' del eps strt ac) =
        cutDFA $ DFA sortedStates alph' del' (rtClose nf (sort [ t | (s, t) <- eps, s == strt
            ])) [st' | st' <- sortedStates, intersect st' ac /= []] where
            del' sy ls = rtClose nf ( unionL [del sy l | l <- ls] )
            sortedStates = map sort $ subsequences sts'

singleAccept :: Ord a => DFA a -> ENFA Int
singleAccept d = ENFA sts alph delt ep st acc where
    d' = makeIntDFA d
    newAccept = maximum (states d') + 1
    sts = newAccept : states d'
    alph = alphabet d'
    delt sym t | t == newAccept = []
                | otherwise = [delta d' sym t]
    ep = [(f, newAccept) | f <- acceptstate d']
    st = start d'
    acc = [newAccept]

-- Checks whether the reachable states of two DFA's are Isomorphic
brzozowski :: (Ord a, Ord b) => DFA a -> DFA b -> Bool
brzozowski d d' = alphabet d == alphabet d' &&
    (null (acceptstate m) && null (acceptstate m')) ||
    isIsomorphTo (start m) (start m') [] where
        m = minimizeDFA d
        m' = minimizeDFA d'
    isIsomorphTo s s' is =
        (s,s') `elem` is || (s `elem` acceptstate m) == (s' `elem` acceptstate m') &&
        all (\sym -> isIsomorphTo (delta m sym s) (delta m' sym s') ((s,s'):is)) (alphabet m)

```

8 Simple Tests

This section discusses our testing suite. We import all modules and test different aspects of the code via several QuickCheck queries.

```

{-# LANGUAGE ScopedTypeVariables #-}
module Main where
import Test.QuickCheck
import RegExp
import DFA
import ENFA
import NFA
import Translation

```

The main function runs all of our tests and prints a one-line explanation for them.

```

main :: IO()
main = do
    print "ZeroStart accepts words starting with 0, for DFA, NFA and ENFA version:"
    test1FA
    print "ZeroStart does not accept other words, for DFA, NFA and ENFA version:"
    test2FA
    print "The reverse of the reverse accepts and rejects the same words:"
    test1ReverseDFA

```

```

print "The reverse of the reverse is isomorphic to the original:"
test2ReverseDFA
print "Minimize yields an automaton that accepts and rejects the same words:"
test1MinimizeDFA
print "Minimize doesn't yield more states:"
test2MinimizeDFA
print "cutDFA DFA accepts and rejects the same as DFA:"
test1CutDFA
print "Powerset construction yields equivalent DFA for NFA:"
test1PowNF
print "Powerset construction yields equivalent DFA for ENFA:"
test1PowENF
print "DFA accepts words generated by the corresponding RegExp:"
test1DFtoReg
print "DFA does not accept words generated by the RegExp corresponding to the
      complement DFA:"
test2DFtoReg
print "ENFA corresponding to RegExp accepts words generated from RegExp:"
test1RegtoENF
print "Composition of translations yields equivalent DFA for DFA:"
test1DFeqv
print "Complement RegExps don't generate the same word:"
test1CompRegx
print "Complement on different RegExp:"
test2CompRegx
print "Minimised automata generated from a RegExp accept words from said RegExp:"
testMinimizeMore

```

First we start by defining some example NFA-s and ϵ -NFA-s similar to the previously seen DFA called zeroStart, that accept the same language. We also add a further example for a DFA.

```

zeroStartNF:: NFA Int
zeroStartNF = NFA [0,1,2, 3] ['0', '1'] deltazeroNF 0 [1] where
  deltazeroNF:: Symbol-> Int -> [Int]
  deltazeroNF char st
    | st == 0 && char == '0' = [1]
    | st == 0 && char == '1' = [2, 3]
    | st == 1 = [1]
    | st == 2 = [2, 3]
    | st == 3 = [3]
    | otherwise = [-1] --added to avoid hlint warning 'pattern matches are non-exhaustive'

zeroStartENF:: ENFA Int
zeroStartENF = ENFA [0,1,2, 3] ['0', '1'] deltazeroNF [(3,2), (2,3)] 0 [1] where
  deltazeroNF:: Symbol-> Int -> [Int]
  deltazeroNF char st
    | st == 0 && char == '0' = [1]
    | st == 0 && char == '1' = [2, 3]
    | st == 1 = [1]
    | st == 2 = [2, 3]
    | st == 3 = [3]
    | otherwise = [-1]

dfa2:: DFA Int
dfa2 = DFA [0,1,3] "01" del2 3 [0] where
  del2 char st
    | char == '0' && st == 0 = 3
    | char == '0' && st == 1 = 0
    | char == '0' && st == 3 = 3
    | char == '1' && st == 0 = 1
    | char == '1' && st == 1 = 3
    | char == '1' && st == 3 = 3
    | otherwise = -1

```

Note the regular expression for the language of words starting with 0 is $0(0 + 1)^*$. We generate words from this expression and test if all 3 of our example automata for this language accept the generated words. Next, we generate words from the complement language, which has the regular expression $\epsilon + 1(0 + 1)^*$ and test if the automata reject these words.

```

test1FA :: IO ()
test1FA = quickCheck (forAll (generateString (Con (R "0") (Star (Union (R "0") (R "1")))))
    (\w -> zeroStart 'evaluate' w && zeroStartNF 'evaluateNF' w &&
        zeroStartENF 'evaluateENF' w ))

test2FA :: IO ()
test2FA = quickCheck (forAll (generateString (Union Epsilon (Con (R "1") (Star (Union (R "0"
    "1") (R "1"))))))
    (\w -> not (zeroStart 'evaluate' w) && not (zeroStartNF 'evaluateNF'
        ' w) && not(
            zeroStartENF 'evaluateENF' w )))

```

Here we test the toolkit for obtaining the minimal DFA for a language. Test if the reverse of the reverse DFA accept the same words as the original DFA. We go further and use the brozowski function to test if they are isomorphic. We then test if a DFA and its corresponding minimal DFA accept the same words. We finish by testing that the minimal DFA has always at most as many states as the original DFA. We also test the cutDFA function. We check if the DFA after cutting accepts the same words as the original DFA. We generate arbitrary DFAs for these.

```

test1ReverseDFA :: IO ()
test1ReverseDFA = quickCheck (\r (d :: DFA Int) -> forAll (generateString r) (\w -> d '
    evaluate' w == (reverseDFA . reverseDFA) d 'evaluate' w ))
test2ReverseDFA :: IO ()
test2ReverseDFA = quickCheck (\(d :: DFA Int) -> brzowski d ((reverseDFA . reverseDFA) d)
    )
test1MinimizeDFA :: IO ()
test1MinimizeDFA = quickCheck (\r (d :: DFA Int) -> forAll (generateString r) (\w -> d '
    evaluate' w == minimizeDFA d 'evaluate' w ))
test2MinimizeDFA :: IO ()
test2MinimizeDFA = quickCheck (\ (d :: DFA Int) -> length (states d) >= length (states (
    minimizeDFA d)))
test1CutDFA :: IO ()
test1CutDFA = quickCheck (\r (d :: DFA Int) -> forAll (generateString r) (\w -> d 'evaluate
    ' w == cutDFA d 'evaluate' w ))

```

We now test the powerset construction. We test if an NFA accepts the same binary strings as its powerset construct. We repeat for ϵ -NFAs. We generate arbitrary NFAs and ϵ -NFAs for this.

```

test1PowNF :: IO ()
test1PowNF = quickCheck (forAll (generateString (Union (R "0") (R "1"))
    (\w (enf :: NFA Int) -> evaluateNF enf w == evaluate (nfaToDFA
        enf) w) )

test1PowENF :: IO ()
test1PowENF = quickCheck (forAll (generateString (Union (R "0") (R "1"))
    (\w (enf :: ENFA Int) -> evaluateENF enf w == evaluate (
        enfaToDFA enf) w) )

```

We now test the translation from DFA to RegExp. We check whether a DFA accepts words generated from the corresponding regular expression and that it does not accept words generated from the regular expression corresponding to the DFA with complement accept states. We generate arbitrary DFAs for this.

```

test1DFtoReg :: IO ()
test1DFtoReg = quickCheck (\ (df :: DFA Int) -> if dfaToRegExp df /= Empty then
    forAll (generateString (dfaToRegExp df)) (\w -> df 'evaluate' w) else
        property True)

test2DFtoReg :: IO ()
test2DFtoReg = quickCheck (\ (df :: DFA Int) -> if dfaToRegExp (flipDFA df) /= Empty then
    forAll (generateString (dfaToRegExp (flipDFA df))) (not . evaluate df) )
    else property True)

```

We now test the translation from `RegExp` to ϵ -NFA. We check whether the ϵ -NFA corresponding to a regular expression accepts words generated from the the expression. We generate arbitrary regular expressions for this.

```
test1RegtoENF :: IO ()
test1RegtoENF = quickCheck ( \ (reg:: RegExp) -> forAll (generateString reg)
                             (\w -> regExpToENFA reg 'evaluateENF' w))
```

We put everything together and test whether a DFA and the DFA obtained from translating to regular expressions, then to ϵ -NFA and back to DFA are equivalent. We test it on our two example DFA-s. We also calculate complement regular expressions by translating to DFAs, taking the complement of accept states and translating back to regular expressions. We test such complement expressions don't yield the same words. We test this on 2 toy expressions. We also combine our translations by the minimal DFA toolkit and check that the minimal DFA corresponding to a regular expression accepts the words generated from it.

```
test1DFeqv :: IO ()
test1DFeqv = quickCheck (forAll (generateString (Union (R "0") (R "1"))))
                        (\w -> evaluate dfa2 w == evaluate (func dfa2) w && evaluate zeroStart w
                          == evaluate (func zeroStart) w)) where
  func = enfaToDFA . regExpToENFA . dfaToRegExp

test1CompRegx :: IO ()
test1CompRegx = quickCheck (forAll (generateString reg) (\w -> forAll (generateString (creg
  reg)) (/= w))) where
  creg = dfaToRegExp . makeIntDFA . flipDFA . enfaToDFA . regExpToENFA
  reg = Union Epsilon (Con (R "1") (Star (Union (R "0") (R "1"))))

test2CompRegx :: IO ()
test2CompRegx = quickCheck (forAll (generateString reg) (\w -> forAll (generateString (creg
  reg)) (/= w))) where
  creg = dfaToRegExp . makeIntDFA . flipDFA . enfaToDFA . regExpToENFA
  reg = Con (Star (R "2")) (Con (R "0") (R "0"))

testMinimizeMore :: IO ()
testMinimizeMore = quickCheck (\r -> forAll (generateString r) (\w -> (minimizeDFA .
  enfaToDFA . regExpToENFA) r 'evaluate' w))
```

The output of the test suite is as follows.

```
"ZeroStart accepts words starting with 0, for DFA, NFA and ENFA version:"
+++ OK, passed 100 tests.
"ZeroStart does not accept other words, for DFA, NFA and ENFA version:"
+++ OK, passed 100 tests.
"The reverse of the reverse accepts and rejects the same words:"
+++ OK, passed 100 tests.
"The reverse of the reverse is isomorphic to the original"
+++ OK, passed 100 tests.
"Minimize yields an automaton that accepts and rejects the same words:"
+++ OK, passed 100 tests.
"Minimize doesn't yield more states:"
+++ OK, passed 100 tests.
"cutDFA dfa accepts and rejects the same as dfa:"
+++ OK, passed 100 tests.
"Powerset construction yields equivalent DFA for NFA:"
+++ OK, passed 100 tests.
"Powerset construction yields equivalent DFA for ENFA:"
+++ OK, passed 100 tests.
"DFA accepts words generated by the corresponding RegExp:"
+++ OK, passed 100 tests.
"DFA does not accept words generated by the RegExp corresponding to the complement DFA:"
+++ OK, passed 100 tests.
"ENFA corresponding to RegExp accepts words generated from RegExp:"
+++ OK, passed 100 tests.
"Composition of translations yields equivalent DFA for DFA:"
+++ OK, passed 100 tests.
"Complement RegExps don't generate the same word:"
+++ OK, passed 100 tests.
"Complement on different RegExp:"
+++ OK, passed 100 tests.
```

```
"Minimised automata generated from a RegExp accept words from said RegExp:"  
+++ OK, passed 100 tests.
```

9 Conclusion

Our primary sources of reference are [Chi18] and [Sip13].

References

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